# Операционные системы

## Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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# 1. Цели и задачи работы

# Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

# Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

# 2. Процесс выполнения лабораторной работы

# Перенаправление ввода-вывода

```
sipolunin@sipolupin:~$ ls /etc/ > file.txt
sipolunin@sipolupin:~$ ls >> file.txt
sipolunin@sipolupin:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

**Рисунок 1:** Запись в файл

# Работа фильтра

```
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$ grep .conf file.txt > conf.txt
sipolunin@sipolupin:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

**Рисунок 2:** Поиск расширения .conf

# Команда поиска

```
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage02/report/bib/cite.bib
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage03/report/_resources/csl
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage03/report/bib/cite.bib
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage04/report/_resources/csl
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage04/report/bib/cite.bib
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage05/report/_resources/csl
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage05/report/bib/cite.bib
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage06/report/_resources/csl
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage06/report/bib/cite.bib
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
esentation/report/_resources/csl
/home/sipolunin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
esentation/report/bib/cite.bib
/home/sipolunin/git-extended/.git/hooks/commit-msg.sample
/home/sipolunin/git-extended/.git/objects/dd/c52e9de0d6a36ac7c909e22398eaec1b78be05
/home/sipolunin/git-extended/.git/objects/cd
/home/sipolunin/git-extended/.git/config
/home/sipolunin/conf.txt
sipolunin@sipolupin:~$
```

**Рисунок 3:** Поиск файлов

# Команда поиска

```
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
/etc/systemd/system/happd.service
```

**Рисунок 4:** Поиск файлов

```
sipolunin@sipolupin:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$ find ~ -name "log*" > logfile &
[1] 6760
sipolunin@sipolupin:~$
[1]+  Done                    find ~ -name "log*" > logfile
sipolunin@sipolupin:~$ rm logfile
sipolunin@sipolupin:~$
```

**Рисунок 5:** Фоновый запуск процесса

```
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$ gedit &
[1] 6802
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$ ps | grep gedit
   6802 pts/1    00:00:00 gedit
sipolunin@sipolupin:~$ kill 6802
[1]+  Terminated              gedit
sipolunin@sipolupin:~$
sipolunin@sipolupin:~$
```

**Рисунок 6:** Фоновый запуск и завершение процесса

# Команды df и du

```
DF(1)                          User Commands                          DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of df.  df displays the amount of
       space  available  on the file system containing each file name argument.  If
       no file name is given, the space available on  all  currently  mounted  file
       systems  is shown.  Space is shown in 1K blocks by default, unless the envi-
       ronment variable POSIXLY_CORRECT is set, in which case 512-byte  blocks  are
       used.

       If  an  argument  is  the  absolute  file name of a device node containing a
       mounted file system, df shows the space available on that file system rather
       than on the file system containing the device node.  This version of df can-
       not show the space available on unmounted  file  systems,  because  on  most
       kinds  of  systems doing so requires non-portable intimate knowledge of file
       system structures.

OPTIONS
       Show information about the file system on which each FILE  resides,  or  all
       file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in
              units of 1,048,576 bytes; see SIZE format below

       --direct
 Manual page df(1) line 1 (press h for help or q to quit)
```

**Рисунок 7:** Справка по команде df

# Команды df и du

```
DU(1)                          User Commands                          DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent sizes rather  than device usage; although the apparent
              size is usually smaller, it may be larger due to holes in  ('sparse')
              files, internal fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in
              units of 1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
              produce a grand total

       -D, --dereference-args
              dereference only symlinks that are listed on the command line
 Manual page du(1) line 1 (press h for help or q to quit)
```

**Рисунок 8:** Запуск команды df

# Команды df и du

```
sipolunin@sipolupin:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91645744  62957152  60% /
devtmpfs         4011284        0   4011284   0% /dev
tmpfs            4035660       92   4035568   1% /dev/shm
tmpfs            1614264     1952   1612312   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.s
ervice
tmpfs            4035660     5704   4029956   1% /tmp
/dev/nvme0n1p3 155186176 91645744  62957152  60% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.s
ervice
tmpfs             807132      164    806968   1% /run/user/1071
/dev/sr0         2677920  2677920         0 100% /run/media/sipolunin/Fedora-WS-Live
-43
sipolunin@sipolupin:~$
```

**Рисунок 9:** Справка по команде du

# Команды df и du

```
4       ./git-extended/.git/objects/dd
4       ./git-extended/.git/objects/9f
4       ./git-extended/.git/objects/b1
4       ./git-extended/.git/objects/b3
4       ./git-extended/.git/objects/cd
4       ./git-extended/.git/objects/2b
4       ./git-extended/.git/objects/16
4       ./git-extended/.git/objects/af
4       ./git-extended/.git/objects/91
76      ./git-extended/.git/objects
8       ./git-extended/.git/logs/refs/heads
8       ./git-extended/.git/logs/refs/remotes/origin
8       ./git-extended/.git/logs/refs/remotes
16      ./git-extended/.git/logs/refs
20      ./git-extended/.git/logs
212     ./git-extended/.git
216     ./git-extended
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
790684  .
sipolunin@sipolupin:~$
```

**Рисунок 10:** Запуск команды du

# Команда поиска

```
/home/sipolunin/git-extended/.git/objects/8b
/home/sipolunin/git-extended/.git/objects/09
/home/sipolunin/git-extended/.git/objects/dd
/home/sipolunin/git-extended/.git/objects/9f
/home/sipolunin/git-extended/.git/objects/b1
/home/sipolunin/git-extended/.git/objects/b3
/home/sipolunin/git-extended/.git/objects/cd
/home/sipolunin/git-extended/.git/objects/2b
/home/sipolunin/git-extended/.git/objects/16
/home/sipolunin/git-extended/.git/objects/af
/home/sipolunin/git-extended/.git/objects/91
/home/sipolunin/git-extended/.git/logs
/home/sipolunin/git-extended/.git/logs/refs
/home/sipolunin/git-extended/.git/logs/refs/heads
/home/sipolunin/git-extended/.git/logs/refs/remotes
/home/sipolunin/git-extended/.git/logs/refs/remotes/origin
/home/sipolunin/monthly
/home/sipolunin/reports
/home/sipolunin/reports/monthly
/home/sipolunin/reports/monthly/monthly
/home/sipolunin/ski.plases
/home/sipolunin/ski.plases/equipment
/home/sipolunin/ski.plases/plans
/home/sipolunin/australia
/home/sipolunin/play
/home/sipolunin/play/games
/home/sipolunin/play/games/play
sipolunin@sipolupin:~$
```

**Рисунок 11:** Поиск директорий

# 3. Выводы по проделанной работе

# Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.